



OPJ Girls College F-8/2, Islamabad

First Term EXAMINATION 2026

SUBJECT: Physics

CLASS: XII

Time allowed: 2 hours 30 min

Total Marks Sections B and C: 39

SECTION - B (Marks)

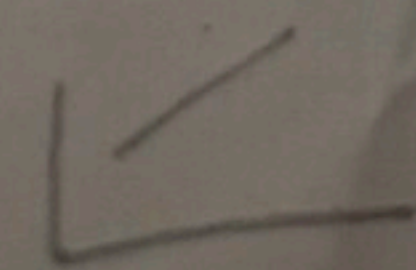
Q.2 Answer the following questions briefly.

(9x3 =27)

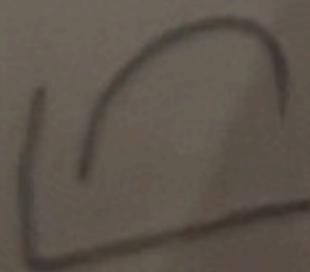
i) Express the following in scientific notation. a) 1168×10^{-27} b) 725×10^5 c) 0.02×10^{-8} <i>1.168 x 10⁻²⁴ 7.25 x 10⁷ 2.0 x 10⁻¹⁰</i>	3	OR	i) A chocolate wrapper is 6.7 cm long and 5.4 cm wide. Calculate its area upto reasonable number of significant figures.	3
ii) Does the speedometer of a car measures its speed or velocity? Give a valid reason to support your answer.	1+2	OR	ii) A person weighs 600N on Earth. What would be his weight on the Moon, where acceleration due to gravity is 1.62 m/s^2 . <i>972N</i>	3
iii) Differentiate between distance and displacement.	3	OR	iii) A car weighing 1200 kg enters a roundabout with diameter of 60m at a speed of 25 km/h. Calculate the centripetal force acting on the car.	3
iv) What is terminal velocity? Plot v-t graph of a paratrooper coming down with terminal velocity.	1+2	OR	iv) Why is it safe to wear helmet while driving motor cycle? Support your answer with a mathematical relation.	2+1
v) If a liquid has a density three times the density of mercury, what will be the height of liquid column in barometer?	3	OR	v) Differentiate between Rolling friction and Sliding friction.	3
vi) Why water tanks are constructed at the highest level in our houses?	3	OR	vi) Calculate the torque acting on a spanner of length 20cm by applying 50N force? What should be the length of spanner to loosen the same nut by 100N force? <i>0.1m</i>	1.5+1.5
vii) A hydraulic lift has 0.002 m^2 narrow cylinder area while 0.9 m^2 wider cylinder area. How much force must be applied if a car weighing 1800 kg is to be lifted?	3	OR	vii) Convert the following. a) 170 km/h into m/s <i>47.2</i> b) 1 m/s^2 into km/h^2 <i>12960</i> c) 36 m/s into km/h <i>129.6</i>	3
viii) Write the conditions for maximum minimum work done.	1.5+1.5	OR	viii) Why does a girl standing in a school bus fall forward when sudden brakes are applied?	3
ix) Differentiate between theory and law with 2 examples each.	3	OR	ix) Plot d-t graph showing a) body moving with constant speed. b) body moving with decreasing speed. c) body at rest.	3

in. of

(a)



(b)



SECTION - C (Marks 12)

Note: Attempt all questions. Marks of each question are given. (6* 2=12)

Q.3.Explain fundamental forces in nature in detail.	6	OR	Q.3.A car travels a curvy track of length 800 m in 40s. The straight path is about 600 m between starting and ending points, which the car travels in 36 s. Find average speed and average velocity of the car.	6
Q.4.Two masses 250 g and 100 g are hanging at positions 65 cm and 80 cm respectively on a uniform rod pivoted at 50 cm. Where should a third mass of 400 g be positioned to balance the rod?	6	OR	Q.4.Differentiate between Accuracy and Precision.	6

speed :- 20 m/s
velocity :- 16.67 m/s

$$x = 16.875 \text{ cm}$$

Km/hr to m/s

$$\times \frac{1000}{3600}$$

m/s to Km/hr

$$\times \frac{3600}{1000}$$

$$d_1 = 65 - 50 = 15 \text{ cm}$$

$$m_1 = 250 \text{ g}$$

$$d_2 = 80 - 50 = 30 \text{ cm}$$

formula :- $m_3 d_3 = m_1 d_1 + m_2 d_2$

let d_3 be x :-

$$400x = (250 \times 15) + (100 \times 30)$$